

CS 2XY3 Midterm Exam (Problems)

Time: 90 minutes. Total: 60 points. Answer all questions.

Question 1 (10 points) - OS Fundamentals

- (a) List three responsibilities of an operating system. (3 pts)
- (b) Compare and contrast monolithic kernels and microkernels. Give one advantage and one disadvantage of each. (4 pts)
- (c) What is a system call? Give two examples. (3 pts)

Question 2 (15 points) - Process Scheduling

Five processes arrive at time 0 with the following burst times: $P_1=10$, $P_2=3$, $P_3=7$, $P_4=1$, $P_5=5$.

- (a) Calculate average waiting time for SJF scheduling. (5 pts)
- (b) Calculate average waiting time for Round Robin with quantum=4. (5 pts)
- (c) Does Round Robin always outperform FCFS? Justify your answer. (5 pts)

Question 3 (15 points) - Memory

A system has 32-bit virtual addresses, 4-KB pages, and 512 MB of physical RAM.

- (a) How many virtual pages exist? (3 pts)
- (b) How many physical frames exist? (3 pts)
- (c) If 8 bits per page table entry are used for protection/status, how large is a single-level page table? (4 pts)
- (d) Why might a multi-level page table be preferred? (5 pts)

Question 4 (20 points) - Synchronization

- (a) What is a race condition? Give a concrete example with two threads incrementing a shared counter. (5 pts)
- (b) Implement a solution to the bounded-buffer problem using semaphores. Show pseudocode for both producer and consumer. (10 pts)
- (c) What is priority inversion? Describe one technique to mitigate it. (5 pts)